

Status

39 / 40
Total Score

97.50%
Percentage

Passed
Result

15
Time Spent (mins)

Question 1 Of 30

Select Question

Q16

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which code snippet correctly creates a scatter plot?

☐ plt.scattergraph([1,2,3],[4,5,6])

☒ plt.scatter([1,2,3],[4,5,6])

✓ Correct Answer

☐ plt.pointplot([1,2,3],[4,5,6])

☐ plt.scatter_plot([1,2,3],[4,5,6])

Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Select Question

Question 2 Of 30

Q10

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which of the following is required when creating a table in SQL?

☒ At least one column must be defined with a data type.

✓ Correct Answer

☐ Table with no columns allowed.

☐ A primary key must be defined.

☐ Table with no datatype allowed.

MacBook Pro

What will the following code print:

```
x = 50
if x < 30:
    print("Low")
elif x == 50:
    print("Exact")
else:
    print("High")
```

☐ Low

☐ Error

☒ Exact

☐ High

✓ Correct Answer

Question 4 Of 30

Select Question

Q21

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

Which SQL query is used to rename the phone column to contact_number in the customers table?

☒ ALTER TABLE customers RENAME COLUMN phone TO contact_number;

✓ Correct Answer

☐ ALTER TABLE customers CHANGE COLUMN phone contact_number VARCHAR(255);

☐ UPDATE customers SET phone=contact_number;

☐ ALTER TABLE customers MODIFY phone contact_number;

Question 5 Of 30

Select Question

Q6

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which SQL snippet correctly uses a CTE to filter sales data for amounts greater than 5000 and then select all columns from the filtered set?

☐ WITH high_sales AS sales WHERE amount>5000;

☒ WITH high_sales AS (SELECT * FROM sales WHERE amount > 5000)
SELECT * FROM high_sales;

✓ Correct Answer

☐ WITH high_sales AS (SELECT * FROM sales WHERE amount > 5000)
SELECT amount FROM sales;

☐ WITH high_sales AS (SELECT * FROM sales WHERE amount > 5000)
SELECT COUNT(*) FROM high_sales;

Score 2

Correct

Type : Mcq

Attempted

What will the following code print:

```
book_available = False
book_reserved = True

if book_available:
    print("Book available for borrowing")
elif book_reserved:
    print("Book reserved")
else:
    print("Book unavailable")
```

☐ Error

☒ Book reserved

☒ Correct Answer

☐ Book unavailable

MacBook Pro

Question 7 Of 30

Select Question

Q24

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

Which query uses a CTE to find the highest salary?

☐ WITH highest_salary AS (SELECT MAX(salary) AS max_salary FROM employees) SELECT * FROM highest_salary;

☐ WITH highest_salary AS (SELECT MAX(salary) FROM employees) SELECT salary FROM employees;

☐ SELECT * FROM highest_salary;

☒ WITH highest_salary AS (SELECT MAX(salary) AS max_salary FROM employees) SELECT max_salary FROM highest_salary;

✔ Correct Answer

Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Select Question

Question 8 Of 30

Q14

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

What happens if an exception is not handled in Python?

- ☐ Program continues
- ☐ Program logs error but continues
- ☐ Program silently exits
- ☒ Program stops with error

✓ Correct Answer

Score

1

Correct

Type : Mcq

Attempted

What will the following code print:

```
x = 50
if x < 30:
    print("Low")
elif x >= 30 and x < 60:
    print("Medium")
else:
    print("High")
```

☐ Error☐ Low☐ High☒ Medium☒ Correct Answer

What will be the output of the following code:

```
for i in range(5):  
    if i == 3:  
        break  
    print(i)
```

☐ 1 2 3 4

☐ 0 1

☒ 0 1 2

☐ 0 1 2 3

☒ Correct Answer

Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Question 11 Of 30

Select Question

Q23

< Previous

Next: >

Score

2

Correct

Type : Mcq

Attempted

A hospital wants to analyze historical patient records to identify patterns. Which system is best?

☐ OLTP System

☐ NoSQL Database

☒ OLAP System

✓ Correct Answer

☐ Operational System

Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Question 12 Of 30

Select Question

Q8

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

Which of the following will raise a ZeroDivisionError in Python?

☐ 0 / 10

☐ 10 / 2

☒ 10 / 0

☐ 0 + 0

✓ Correct Answer

Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Question 13 Of 30

Select Question

Q9

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

What will happen if the condition in an assert statement evaluates to False?

☒ It raises AssertionError

✓ Correct Answer

☐ The program continues

☐ It will print a warning message

☐ It returns False

Question 14 Of 30

Select Question

Q1

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

What is the output of the following code:

```
x = [1, 2, 3]
y = x
y[0] = 10
print(x)
```

☐ Error

☒ [10, 2, 3]

☒ Correct Answer

☐ [5, 2, 3]

Question 15 Of 30

Select Question

Q5

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which SQL query returns all employees who have a salary greater than or equal to 60000 and work in the 'IT' department?

☐ SELECT * FROM employees WHERE salary >= 60000 OR department='IT';

☐ SELECT * FROM employees WHERE salary > 60000 AND department='IT';

☒ SELECT * FROM employees WHERE salary >= 60000 AND department='IT';

✓ Correct Answer

☐ SELECT * FROM employees WHERE salary > 60000 OR department='IT';

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Submitted

Status

39 / 40

Total Score

97.50%

Percentage

Passed

Result

15

Time Spent (mins)

Select Question

Question 16 Of 30

Q15

< Previous

Next: >

Score

1

Correct

Type : Mcq

Attempted

What is the correct way to connect to an SQLite database using Python?

☐ `conn = sqlite3.database('mydb.db')`

☒ `conn = sqlite3.connect('mydb.db')`

✓ Correct Answer

☐ `conn = sqlite3.open('mydb.db')`

☐ `conn = sqlite3.start('mydb.db')`

Score 2

Correct

Type : Mcq

Attempted

You need to display a report of all authors and their books. Authors without any books should appear with NULL values. Which query structure should you use?

☐ SELECT authors.author_id, books.book_id FROM authors INNER JOIN books ON authors.author_id = books.author_id;

☐ SELECT authors.author_id, books.book_id FROM authors FULL JOIN books ON authors.author_id = books.author_id;

☐ SELECT authors.author_id, books.book_id FROM authors RIGHT JOIN books ON authors.author_id = books.author_id;

☒ SELECT authors.author_id, books.book_id FROM authors LEFT JOIN books ON authors.author_id = books.author_id;

✓ Correct Answer

Question 18 Of 30

Select Question

Q26

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

You want to find the top 5 customers based on spending using a CTE. Which query works?

- ☐ SELECT customer_id, SUM(amount_spent) FROM orders GROUP BY customer_id LIMIT 5;
- ☐ WITH customers AS (SELECT * FROM orders) SELECT * FROM orders;
- ☐ SELECT * FROM top_customers ORDER BY total_spent DESC LIMIT 5;
- ☒ WITH top_customers AS (SELECT customer_id, SUM(amount_spent) AS total_spent FROM orders GROUP BY customer_id) SELECT * FROM top_customers ORDER BY total_spent DESC LIMIT 5;

✓ Correct Answer

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Status

39 / 40
Total Score

97.50%
Percentage

Passed
Result

15
Time Spent (mins)

Question 19 Of 30

Select Question

Q18

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

What does the 'as e' syntax do in Python's exception handling?

- ☐ Defines alias for exception
- ☐ Rethrows exception
- ☒ Assigns exception object to variable e
- ☐ Suppresses exception

✓ Correct Answer

MacBook Pro

Status

Total Score

97.50%
Percentage

Passed
Result

15
Time Spent (mins)

Question 20 Of 30

Select Question

Q3

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which SQL statement is used to retrieve unique values from a column?

- ☐ SELECT FILTERED
- ☐ SELECT UNIQUE_VALUES
- ☐ SELECT UNIQUE
- ☒ SELECT DISTINCT

✓ Correct Answer

MacBook Pro

Status

Total Score

97.50%
Percentage

Passed
Result

15
Time Spent (mins)

Select Question

Question 21 Of 30

Q13

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which function is used to create a pie chart in Matplotlib?

☐ plt.plot()

☐ plt.hist()

☐ plt.bar()

☒ plt.pie()

✓ Correct Answer

MacBook Pro

Total Score

Percentage

Passed

Result

15

Time Spent (mins)

Select Question

Question 22 Of 30

Q29

< Previous

Next >

Score 2

Correct

Type : Mcq

Attempted

You have a dataset `df` with a category column and a revenue column. You want to calculate total revenue for each product category. Which operation in Pandas would you use

☐ `df.groupby('revenue')['category'].sum()`

☒ `df.groupby('category')['revenue'].sum()`

✓ Correct Answer

☐ `df['revenue'].sum()`

☐ `df['revenue'].groupby('category').mean()`

MacBook Pro

Question 23 Of 30

Select Question

Q4

< Previous

Next: >

Score 0

Incorrect

Type : Mcq

Attempted

Which code snippet correctly adds labels to the x and y axes in a Matplotlib plot?

☐ `plt.xlabel('X axis'); plt.ylabel('Y axis')`

✓ Correct Answer

☐ `plt.set_xlabel('X axis'); plt.set_ylabel('Y axis')`☐ `plt.add_labels('X axis','Y axis')`☒ `plt.label('X axis'); plt.label('Y axis')`

✗ Incorrect Answer

MacBook Pro

Score 2

Correct

Type : Mcq

Attempted

You want to display all products and their prices, including products with no price listed. Which query will work?

☐ SELECT products.product_name, prices.price FROM products FULL JOIN prices ON products.product_id = prices.product_id;

☐ SELECT products.product_name, prices.price FROM prices JOIN products;

☐ SELECT products.product_name, prices.price FROM products RIGHT JOIN prices ON products.product_id = prices.product_id;

☒ SELECT products.product_name, prices.price FROM products LEFT JOIN prices ON products.product_id = prices.product_id;

✓ Correct Answer

Score 2

Correct

Type : Mcq

Attempted

You need to find all customers who have placed more orders than the average number of orders placed by customers. Which query correctly uses a subquery?

☐ SELECT * FROM customers WHERE order_count < (SELECT AVG(order_count) FROM customers);

☐ SELECT * FROM customers WHERE order_count >= (SELECT AVG(order_count) FROM customers);

☒ SELECT * FROM customers WHERE order_count > (SELECT AVG(order_count) FROM customers);

☒ Correct Answer

☐ SELECT * FROM customers WHERE order_count = (SELECT AVG(order_count) FROM customers);

Score 1

Correct

Type : Mcq

Attempted

Which query returns all employees whose salary is greater than the salary of employee with id 5?

☐ SELECT * FROM employees WHERE salary < (SELECT salary FROM employees WHERE employee_id=5);

☐ SELECT * FROM employees WHERE salary = (SELECT salary FROM employees WHERE employee_id=5);

☐ SELECT * FROM employees WHERE salary >= (SELECT salary FROM employees WHERE employee_id=5);

☒ SELECT * FROM employees WHERE salary > (SELECT salary FROM employees WHERE employee_id=5);

✓ Correct Answer

Correct

Type : Mcq

Attempted

What will be the output of the following code:

```
try:  
    x = int('abc')  
except ValueError:  
    print('Invalid integer value')
```

☐ Nothing

☐ Error

☒ Invalid integer value

☐ Invalid

✓ Correct Answer

Question 28 Of 30

Select Question

Q19

< Previous

Next: >

Score 1

Correct

Type : Mcq

Attempted

Which SQL query returns the maximum salary in department 5?

- ☐ SELECT salary FROM employees WHERE dept_id=5;
- ☐ SELECT MAX(*) FROM employees;
- ☐ SELECT MAX(salary) FROM employees;
- ☒ SELECT MAX(salary) FROM employees WHERE dept_id=5;

✔ Correct Answer

MacBook Pro

Select Question

Question 29 Of 30

Q22

< Previous

Next: >

Score 2

Correct

Type : Mcq

Attempted

Which data architecture is specifically designed for analytical insights focused on a particular department or business function?

- ☐ Data Lake
- ☐ Data Warehouse
- ☒ Data Mart
- ☐ Operational Database

✓ Correct Answer

MacBook Pro

Score

1

Correct

Type : Mcq

Attempted

What will be the output of the following code:

```
nums = [10, 20, 30, 40, 50]  
print(nums[-3:-1])
```

☐ [20, 30]

☐ [10, 30]

☒ [30, 40]

☐ [40, 50]



Correct Answer